

Yes, and that's a good observation.

The roadmap you pasted is the roadmap I recommended earlier. Since then, we've gone deeper into the topics, so I'd update it slightly based on what you've already completed.

Your Current Progress

Phase 1 – Traditional NLP (Completed)

-  Text Preprocessing
 - Normalization
 - Tokenization
 - Stop Word Removal
 - Stemming
 - Lemmatization
 - Punctuation Removal
-  N-Grams
-  Bag of Words
-  TF
-  IDF
-  TF-IDF
-  Cosine Similarity

Projects you can already build:

-  FAQ Search Engine
-  Document Search
-  Resume Search
-  Similar Document Finder

Phase 2 – Machine Learning (You have already covered much of this)

From our previous conversations, you've already worked on:

-  Spam Detection
-  House Price/Rent Prediction
-  ML.NET basics
-  Naive Bayes (conceptually)
-  Logistic Regression
-  Decision Trees
-  Random Forest (basic)
-  Accuracy
-  Precision

-  Recall
-  F1 Score
-  Confusion Matrix

Projects:

-  Spam Detection
 -  Sentiment Analysis (recommended)
 -  Intent Classification (recommended)
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Phase 3 – Modern NLP (Current Phase)

Completed:

-  Word2Vec
-  One-Hot Encoding
-  Embedding Layer
-  Sentence Embeddings

Still to learn:

-  GloVe
-  FastText
-  Transformer
-  Attention Mechanism
-  BERT
-  GPT
-  LLM Fundamentals

This is where we are now.

Phase 4 – Vector Search

We'll cover:

- Embeddings
- Cosine Similarity on embeddings
- Vector Search
- ANN (Approximate Nearest Neighbor)
- Vector Databases
 - Chroma
 - Qdrant
 - Milvus
 - Pinecone

Then you'll build:

- Semantic Search Engine

Phase 5 – RAG

You'll build a complete RAG pipeline:

PDF

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Chunk

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Embedding

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Vector DB

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Retrieve

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LLM

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Answer



Projects:

- Company Knowledge Chatbot
- HR Assistant
- Policy Assistant
- Document QA

Phase 6 – LLM Integration (.NET)

You'll learn:

- OpenAI API
- Azure OpenAI
- Local LLMs
- Prompt Engineering
- Function Calling
- Structured JSON Output
- Streaming
- Conversation Memory

Projects:

- AI Chat Application
- AI Email Generator

- AI SQL Assistant

Phase 7 – AI Agents

You'll learn:

- Planning
- Memory
- Tool Calling
- Multi-step Reasoning
- Multi-Agent Systems

Projects:

- Coding Assistant
 - SQL Assistant
 - Support Agent
 - Email Agent
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The only change I'd make

The original roadmap jumped from **Sentence Embeddings** directly to **Vector Search**.

After working through the concepts with you, I think there are a few foundational topics that fit naturally in between:

Traditional NLP

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Word2Vec

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Sentence Embeddings

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● Neural Networks

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● Attention Mechanism

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● Transformer

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● BERT

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● GPT

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Embedding Models

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Vector Search

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RAG



LLM Integration



AI Agents

Those additional topics explain **how modern embedding models and LLMs actually work**, making the later phases much easier to understand.

My recommendation

Since you're learning for understanding—not just to use APIs—I recommend we continue with this order:

1. **Neural Network Basics** (how AI models learn)
2. **Attention Mechanism**
3. **Transformer Architecture**
4. **BERT**
5. **GPT**
6. **LLM Fundamentals**
7. **Embedding Models**
8. **Vector Search**
9. **RAG**
10. **LLM Integration in C#**
11. **AI Agents**

This sequence builds the concepts progressively, and each topic prepares you for the next. By the time you reach RAG and AI Agents, you'll understand both the underlying theory and how to implement them in .NET.